

WIRELESS TECHNOLOGY For Safety And Security Of The DEEP SEA

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This article discusses the wireless technologies for the safety and security of the deep sea, including ship-to-shore radios, GPS-tracking solutions and RFID transponders

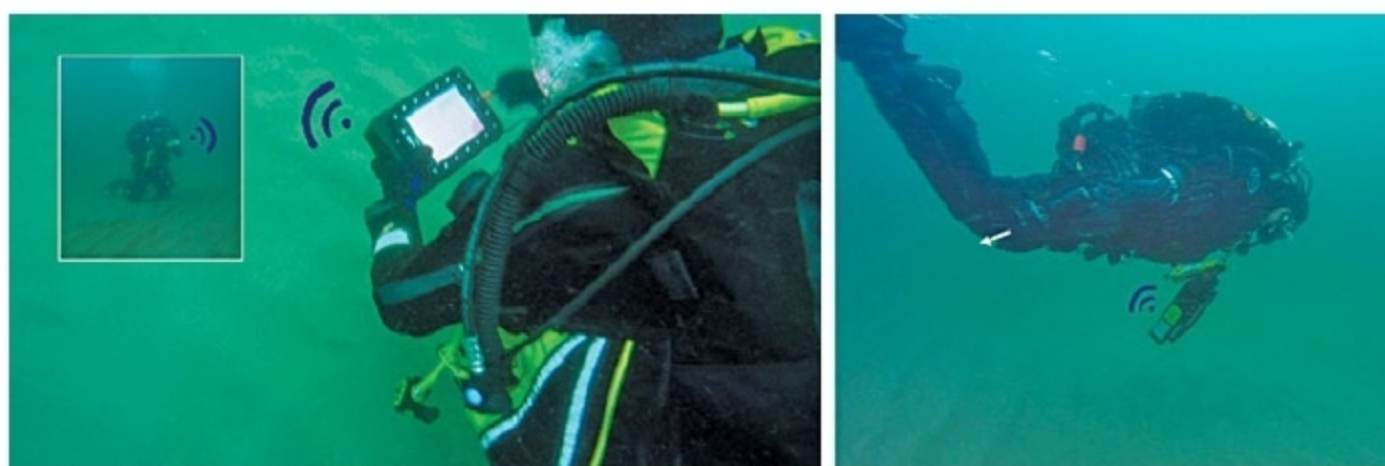


Fig. 1: SWiCOM (Credit: WFS Technologies)

Wireless technologies have made their way into deep sea applications. According to Brendan Hyland, chairman, WFS technologies Ltd, sub-sea Internet of Things (IoT) is a network of smart, wireless sensors and devices configured to provide actionable operational intelligence such as performance, condition and diagnostic information. Sambit Sengupta, associate director - field applications, Avnet India, adds, "The best wireless technology possible today is through a complicated network of satellites and multi-hop systems. Within ships and submarines, infrastructure can support local wireless networks. But with the outside world, there has to be a mixture of acoustics and radio technology with good satellite-based communication."

Underwater wireless communication systems

Ship-to-shore radios, global positioning system (GPS)-tracking solutions, radio frequency identification (RFID) transponders for freight monitoring and marine radars are the various forms of wireless communication used underwater. These extend real-time communication to personnel, vehicles and sensors in the most challenging environments.

Optical signals are used for high-speed, short-range water communication in mild environments. Whereas, acoustics are used for long-range communication in controlled environments. There must be a combination of radio and acoustic technologies to achieve efficient wireless communication under deep sea.

For example, Seatooth Hybrid is a wireless underwater communication system that uses optical, acoustic and RF electromagnetic carrier signals. Its features are:

- Acoustic data rates: 500bps and 1200bps
- Data rates for radio: 500bps, 1.2kbps and 2.4kbps
- Optical data rates: 10Mbps and 100Mbps
- Depth rate: 200 metres
- Operating temperature: -10°C to +60°C
- Data interfaces: RS232 and RS485

Acoustic transmission. Sound travels far in water and, hence, underwater loudspeakers and hydrophones can cover larger distances. Sonic communication equipment are generally placed in the seabed that is frequently travelled